

Unity requires physically enforced agreement

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Abstract

A conscious scene is experienced as one scene, although many brain regions contribute to it. We argue that this unity requires physically enforced agreement: the regions must be coupled so that any change in one, however small, as long as it exceeds the region's own noise, pulls the others toward agreement within the time a percept takes to form. The condition is functional, but it is stated at the grain of graded physical dynamics rather than at the grain of an algorithm, and at that grain cortex and digital chips come apart. It predicts that when two cortical regions are reconnected by an analog loop and by a quantized loop carrying at least as much information, in total and about the task, unity follows the analog loop. An account stated on the information that regions exchange predicts no difference. We prove that under noise, content read from bits responds to any change below its margin by at most its error rate, a bound that is attained, so a digital machine, clocked or clockless, fails the condition whatever algorithm it runs. Applying one test to both substrates, a noisy cortical membrane, modelled or recorded in awake mice, passes near its firing threshold and reaches such a state within a percept's time scale, while the value a clocked latch reads passes at none of its

states. The condition also bounds how slowly cortical coupling may relax, and on a noisy sheet of coupled oscillators the coupling's reach decides how far agreement holds. We specify an animal preparation for the bridge and the outcomes that would refute the premise.

1 Introduction

When you watch a bird fly across the sky, you see one bird in one scene. Yet many parts of your brain work on that scene at once. A visual area tracks where the bird is, and a parietal area tracks the same location to plan a reach toward it. Each part describes a piece of the same situation, and their descriptions agree. This paper asks what kind of agreement that has to be.

Parts can agree in two ways. Picture first a set of weights joined by springs. Each weight's position is a graded physical quantity that pulls on its neighbours all the time, so no weight can move, even slightly, without dragging the others after it. The weights are coordinates of one physical state, and above their own noise they cannot vary independently at any scale. We call this agreement *physically enforced*. Picture next a set of account books kept consistent by a clerk, who at fixed times compares the books and corrects any mismatch. Between checks a smudge on one page changes nothing in any other book. The books agree only because a rule makes them agree, and we call this agreement *computed*. The springs make one system in the sense physics uses; the books are many systems kept in step by a protocol.

A digital chip is of the second kind. Each bit is a separate record, built to ignore small disturbances: a voltage a little above or below the level for 1 still reads as 1. That makes computers reliable, and it also seals each bit off from small changes in every other bit, so whatever agreement a program produces among its bits is computed. Our premise is that the unity of an experience is agreement of the first kind.

Premise P (physical unity). A system’s contents are unified only if each region’s contents are read from graded carriers and, within every window of the contents’ own time scale, some carrier of each region depends on the graded physical state of every region it overlaps, above the noise floor and through a coupling that contracts their disagreement.

In short, the regions that contribute to one experience must be held together by a physical coupling that feels arbitrarily small changes above noise, and not by an exchange of symbols. P is a necessary condition for unity, and a condition on how a system’s conscious contents are held together: which contents are conscious, and what they are like, lie outside it. It says nothing about where in the brain unified contents live: it constrains whichever regions carry them, whether they span the cortex or are confined to a posterior zone of it (Koch et al., 2016).

A finer grain of function. P is still a functional condition: it picks out unity by what physical parts do to one another. What sets it apart is the level of detail, or *grain*, at which it describes them. Functionalism leaves the grain open. Described coarsely enough, a nation whose citizens exchange messages by radio could run a brain’s program (Block, 1978), while the biology of cognition has been argued to tie mental states to a finer grain (Godfrey-Smith, 2016). P sits at the grain of graded physical dynamics, and computational functionalism about unity at the grain of the algorithm. For cortex the two grains agree. They come apart for a perfect digital simulation of cortex, which is unified at the algorithm’s grain and fails P, since its agreement is kept by a rule acting on separate records. The experiment of Section 6 asks which grain unity tracks, holding fixed the information two brain regions exchange and changing only whether the link between them is graded or digital.

The argument. The case for P rests on two premises, both defended in Section 3.

- *Wholeness*, an account of unity: unified experiences are parts of one total experience, so they need one physical system to carry them.
- *Counting*, how physics counts systems: two parts are one system when each responds to the other at every scale between its own noise and its own thresholds.

Every step between these premises and P is proved. Three results do the work. Content read from bits responds to a small change by at most its error rate. A rule that compares records only adds another record, and it joins none of them into one system. A graded reader does join the regions it reads.

Noise and time. Two refinements make P fit real brains. Physical systems are noisy, so P is read on probabilities: a change counts only if it is larger than the part’s own noise, and it must then move what the part does next by at least as much as the part fluctuates by itself. And a neuron is sensitive to small inputs mainly when it is close to firing, so asking every neuron to be sensitive at every instant would count no network of spiking cells as one system. P asks instead that *some* carrier of each region register the other regions within each window of the content’s time scale.

The paper makes three claims:

1. When two cortical regions are bridged by an analog and by a quantized loop with the quantized loop carrying at least as much information, in total and about the task, P predicts that unity follows the analog loop; a weak, sub-threshold nudge to one region should also shift the graded content decoded in an overlapping region toward agreement (Section 6).
2. Under noise, content read from bits responds to any change below its margin by at most its error rate, a bound that is attained, so no digital machine meets P, whatever algorithm it runs; by one test applied to both, a cortical membrane, modelled or recorded in awake mice,

passes near its threshold within a percept’s time scale, while the value a clocked latch reads passes at none of its states (Section 3).

3. Enforced agreement must arrive before the content changes, so P predicts that cortical coupling relaxes within 166 ms across the cortex; on a noisy sheet of coupled oscillators, the coupling’s reach decides how far agreement in phase holds (Section 4).

Every formal result is machine-checked in Lean 4 (Appendix K). The two premises are argued, the proofs run from them, and the bridge tests them together.

2 Unity as agreement between overlapping regions

Think of an atlas. Each page maps one patch of ground, neighbouring pages overlap, and where they overlap they show the same roads. Pages that agree on every overlap fit together into one map of the whole territory. We treat the brain’s regions the same way. Each describes part of a shared situation, and unity is the property that these local descriptions agree wherever they overlap. When they do, they fix a single global state.

Synchrony is not unity: two regions can oscillate in phase while encoding a stimulus at two different locations. Unity is about what overlapping regions describe, and synchrony is at most one route to it.

Carriers and traces. A region’s *content* is a variable the system itself reads, not merely one an outside observer can decode: the regions it drives respond to it, and it determines the quantity the region represents, such as a location. A variable that meets both conditions is a *carrier*; one that only an observer can decode is a *trace*. The supply current of a digital circuit reveals enough about its logical state to recover a secret key (Kocher et al., 1999), yet

no gate reads it, so it is a trace. In cortex, membrane potentials and spike times are carriers, because the cells they drive integrate them. The same test says *when* a variable is a carrier: at the moments its readers take it in. Feed-forward inhibition confines a pyramidal cell’s integration of its inputs to a window of a few milliseconds (Pouille & Scanziani, 2001; Gabernet et al., 2005), short against the interval between the spikes it receives (König et al., 1996), and the timing of first spikes in somatosensory cortex carries information about a stimulus that the spike count does not (Panzeri et al., 2001). A clocked latch, by contrast, looks at its input only during a short aperture at the clock edge. The input’s value there is a carrier. The course of the voltage between edges, which nothing reads, is a trace, like the supply current. So a carrier’s *next content* is what its readers take from it next, at the time they take it: for a membrane, the time of its next spike; for a node that feeds a latch, its value at the next clock edge.

Agreement is measured between regions whose contents represent a shared variable. The regions, and each one’s content variable, are fixed by this causal role and by decodability before any agreement is measured. Nobody can make the measurement trivial by choosing the regions after the fact.

A running example. A visual area represents the bird’s location on a retinotopic map, and a parietal area represents the same location to plan a reach. Each area’s content is its estimate of the location, carried by the membrane potentials and spike times of the cells that encode it, and unity, here, is the two estimates agreeing.

The definition, stated precisely. Region i carries a content s_i , a description of the variables it represents, and regions i and j overlap in the variables both represent. Restricting a content to an overlap keeps only the shared variables. The contents are *compatible* when every two agree on their overlap, $s_i|_{i \cap j} = s_j|_{i \cap j}$. Compatible contents extend to exactly one global description whose restriction to each region is that region’s content. Unity is

compatibility, and its degree is the *overlap discrepancy*: the largest distance between two contents restricted to a common overlap.

Approximate agreement still fixes one description: if every overlap discrepancy of real-valued contents is at most ε , a weighted average of the contents that cover each variable is within ε of every region’s content. Agreement is measured by decoding, and two accurate decoders agree whatever produced their agreement, since $|d_A - d_B| \leq |e_A| + |e_B|$ for errors e_A and e_B . The measure is therefore the agreement *in excess of* what the two errors imply. These results are proved in a companion paper (Bobadilla-Suarez, 2026), and P adds a condition on how that agreement is produced.

Two stages. The companion proposes agreement on overlaps as a correlate of unity, to be tested, and P is conditional on that identification. Its test asks whether excess decoded agreement is larger on trials the subject perceives than on trials it does not. A positive result makes the bridge of Section 6 the next question; a negative one removes decoded agreement as a neural measure of unity and leaves the bridge its behavioural readout, so the second stage does not wait on the first.

3 Enforced agreement and computed agreement

The argument in outline. Five definitions make “physically enforced” precise. Two theorems show that content read from bits does not respond to small changes, without noise and with it. Two premises, wholeness and counting, turn that fact into a claim about unity, and the rest is proved. One measurement protocol, applied to a cortical membrane and to a logic node, shows where each substrate holds its carriers.

Definitions. Each is stated in words first.

- *Micro state and content.* A system has a physical state, its *micro state* $x = (x_i)_i$, made up of the states x_i of its regions. One step of its physics carries x to $f(x)$. A *content map* π reads each region's content off the micro state. The content level is *closed* when the next content can be computed from the current content alone, without looking at the micro state: $\pi \circ f = g \circ \pi$ for some content dynamics g . A digital computer's logical level is closed by design. The next content $\pi(f(x))$ of a carrier is what its readers take from it next.
- *Noise margin.* A readout has a noise margin on a set of states when each state in it has a neighbourhood on which the readout is constant, as any voltage well above a bit's threshold reads as 1.
- *Graded dependence.* Content depends gradedly on a region when every change to that region, however small, can move the next content. Formally, no neighbourhood of the region's micro state leaves the next content unchanged while the other regions are held fixed.
- *Contraction.* The overlap discrepancy *contracts* near a state when one step of f shrinks it by a factor below one throughout a neighbourhood of that state, so perturbations too small to cross any threshold are pulled back toward agreement along with larger ones.
- *Physical enforcement.* Agreement is physically enforced at a state when content depends gradedly on each region there and the overlap discrepancy contracts near it. P requires this at the states the system occupies while its contents are unified.

The margin lemma. In words: content read through a margin cannot feel small enough physical changes. Precisely, if the micro dynamics is continuous and carries a state into a margin's operating set, the next content depends gradedly on no region's micro state. The proof takes one line: continuity

carries a small enough neighbourhood of each region’s micro state into the neighbourhood on which the readout is constant. A finite word of thresholded quantities has a margin away from its thresholds, so the lemma covers any content read from bits. It is elementary by design: a reader who disputes the separation disputes a definition, and the rest of this section argues for each.

Noise. A gate’s chance of flipping is a smooth function of its input voltage and never exactly zero, so the *probability* of a digital machine’s next content depends, very slightly, on every region, and the exact margin lemma would mark a difference below the noise of both substrates. P is therefore read on probabilities.

The *response* of a content to a region at scale η is the largest change that moving the region by at most η makes to the probability of any outcome of the next content, a total-variation distance. The yardstick for a large response is the response that a shift of one standard deviation produces in a Gaussian carrier, $2\Phi(1/2) - 1 = 0.38$, and call it *the fluctuation*: a response this large moves the content by as much as the content naturally fluctuates. Graded dependence above the noise floor asks the response to reach the fluctuation for every change of the region that is larger than its own noise and smaller than the change that crosses one of its thresholds. When the region moves a carrier linearly under additive noise whose density has a single peak, Gaussian or heavy-tailed, a larger change moves the content at least as much as a smaller one, and the criterion is met exactly when a change of one noise amplitude reaches the fluctuation (Appendix C treats noise whose law changes with the state). No constant is fitted to either substrate.

The noisy margin theorem. The margin lemma has a noisy form, and it is this form that carries the argument. Let each realization of the noise fix an L -Lipschitz step of the micro dynamics, one that stretches no distance by more than a factor L . Let the content be constant within radius r of the noisy successor, except on a set of realizations of probability at most ε .

Then a change of one region smaller than r/L moves the probability of every outcome of the next content by at most ε . For anything computed from bits, ε is the probability that some bit of the noisy successor lands within δ of its threshold, and the bound is attained. In plain terms: below its margin, a bit feels a change only through its error rate.

The simulation objection. Why state P on the physical state, below the level of contents? Because at the content level a computer can imitate a coupled field exactly: a digital system that integrates the field's equations has a content map whose next step shrinks the disagreement between overlapping regions just as the field does. P is stated instead on how content depends on graded physical state, and there the two differ. The field's content moves with every graded change in the regions it overlaps; the simulation's content is read from bits held inside their margins and moves with none of them. So a perfect simulation of a coupled cortical field reproduces its content-level trajectory and fails P. The next paragraphs give the reason for that verdict.

Why unity tracks enforcement. Unity settles how many subjects there are: a unified experience is had by one subject and a divided one by two or more. So whatever makes contents unified also decides whether they belong to one system or to several, and agreement at the content level cannot settle that count. Put the records of one program in different buildings, stretch the clerk's step to a year, or let one processor visit the records in turn. The content trajectory stays the same, while the system that realises it goes from one chip to several machines to one record at a time. Block's nation, whose citizens run a brain's functional organisation by radio (Block, 1978), is the same program spread over a country. A criterion stated on content-level agreement gives all of these the same number of subjects, with no physical fact to consult.

Wholeness: unity is not one more content. Wholeness takes experiences to be unified when they are parts of a single total experience, the subject’s experience at that time (Bayne, 2010). Being a part of a whole is not a further part of it. So the unity of two contents is not a third content alongside them, and that limits what can realise it.

Records join nothing. A *record* is a content held by a margin, sealed off from the graded state of everything else. A rule that compares records and corrects them reads and writes records, and whatever it produces is one more record. Once systems are counted as the counting premise below counts them, by chains of couplings at the noise floor, three facts follow as theorems.

- A region whose content is held by a margin, with an error rate below the fluctuation, is coupled to no region that admits a change above its noise inside the margin, whatever the regions it reads carry. It is one system only with itself.
- For any set of records, records that read records included, two regions are one system exactly when a chain of couplings that avoids every record joins them. Records drop out of every chain, so rules that check rules add nothing at any depth.
- Writing corrections back changes nothing. When one region’s next content depends on another only through a record’s symbol, a change of the other inside the margin moves it by at most the record’s error rate, so the rule couples the two no more than the record does.

Graded readers join. A graded reader stays in the chain. A region whose content moves with two regions above their noise, and moves them in turn, joins them into one system even when they share no coupling of their own. Under linear dynamics with uniform noise such a reader joins two regions that a record reading the same two leaves apart (Appendix K). The regress

stops at a reader on the chain, not at a record beside it. A register holding the bird's location as a number is a record; a cell in a third area that integrates spike times from the visual and the parietal area, and projects back to both, is a reader. A workspace that makes every record available to a common reader is the rule again. Computation adds contents and never produces the whole they are parts of. What can make contents parts of one whole, without adding a content, is a physical state whose coordinates carry them, and whose agreement is a fact about where that state sits, written nowhere. P asks for exactly this.

Counting: systems at the noise floor. Physics counts systems by interaction: parts form one system when their joint state does not split into independent coordinates. At no particular scale this makes every chip one system, through wires, supply and crosstalk, which is correct for the voltages. But unity is a property of contents, a chip reads its contents through margins, and whether two coordinates split depends on the resolution at which they are taken. Three steps fix that resolution from the parts themselves.

1. *The lower end is the part's own noise.* A difference smaller than a part's noise does not survive the part's next step, so it is not a state the part holds, nor a coordinate along which it can be independent of anything.
2. *The upper end is the part's nearest threshold.* A change that crosses a threshold changes the part's symbol, and what another part registers of the crossing is the symbol, the same whatever graded state produced it. That registration is a message, an exchange of records, and records join nothing.
3. *In between, physics asks for a product.* Two parts are separate systems at a resolution when the dynamics of each is independent of the other's state at that resolution. A record meets this outright, since no change

inside its margin moves it beyond its error rate. A graded reader is the opposite case: every change of the region it reads, between noise and threshold, moves it by at least its own fluctuation.

The count takes the strict reading of the third step. Two carriers are one system when *every* change of one between its noise and its nearest threshold moves the other by at least the other's own fluctuation, and two when *some* such change leaves the other unmoved. A product decomposition asks less: it may join a carrier that registers some changes in the range and not others. The two agree on records and on graded readers, the two cases the argument turns on, and between them the strict reading is the first two steps applied inside the range. Where a larger change moves the content at least as much as a smaller one, as it does under the single-peaked noise treated above, a carrier can miss a change in the range in two ways. It can miss the small changes and register the large ones, which is a margin narrower than the range. Or it can register changes on one side of the state and miss those on the other, which is a threshold at the state, and then what it registers is which side was taken, a symbol. Both are the ways a record registers, so the strict reading adds no condition of its own. Both ends of the range belong to each part, the size of its noise and the position of its thresholds, so nobody chooses the scale to separate brains from chips.

A precedent: clusters in a gas. Physics already counts systems by comparing an interaction with the parts' own fluctuations. In the theory of imperfect gases, two molecules belong to one physical cluster when their attraction outweighs the kinetic energy of their relative thermal motion, and a cluster is a set of molecules joined by chains of such bonds (Hill, 1955; Sator, 2003). Molecules whose attraction is weaker than their thermal motion still interact, and that interaction changes the pressure of the gas, yet it joins them into no cluster. The criterion has the three features of the count: the scale is set by the parts' own noise, systems are joined through chains, and a

real interaction below that scale leaves the parts separate, as crosstalk below a bit's margin leaves bits separate.

From systems to P. The count is the noise-floor reading of graded dependence given above, and the identity is proved: for carriers whose contents overlap, each depends gradedly above the noise floor on every carrier it overlaps exactly when every overlapping pair is one system. A region holds many carriers, and a noisy carrier registers another only while it sits near its own threshold, as measured below. So two regions are one system when some carrier of each registers some carrier of the other, and P asks this within every window of the contents' time scale. Asking it of some carriers is strictly weaker than asking it of all: a state exists at which the first holds and the second fails.

The chip, counted. A logic node of 1 fF at room temperature has thermal noise of 2 mV (Johnson, 1928; Nyquist, 1928), and the textbook inverter's margin is 506 times that. A change of one noise amplitude flips the bit with probability below 10^{-100} (Kish, 2002), so by the noisy margin theorem no bit registers any other, and each bit is a system of its own. Between thresholds, where a working chip spends nearly all of its time, its contents are as many systems as it has bits, carried on voltages that form one (Appendix D).

Where the line falls. A criterion built to exclude chips would draw its line between substrates. This one draws it inside each substrate, at a carrier's distance from its own threshold, and it discards threshold crossings in cortex as in silicon: the occurrence of a spike is a message, and a spike count latched at a clock edge is a record. Cortex passes on what is left, the range below threshold in which a membrane moves with each input above its noise. An analog crossbar whose column currents drive the next array directly passes, and a neural circuit read only from latched counts fails (Section 5).

The membrane, counted. A spike time passes the test at the moment the membrane rises through threshold, where one unitary cortical input shifts the spike by 1.5 times its jitter (Appendix C). To see how often a membrane is in such a state, the protocol applied to the logic node is applied to a leaky integrate-and-fire membrane driven by white noise, over every state it occupies, with time constants of 5 to 20 ms (Destexhe & Paré, 1999; Destexhe et al., 2003) and rates of 0.5 to 8 Hz (Hromádka et al., 2008). It sits on average 2.0 to 3.5 noise amplitudes below threshold and passes at 0.013 to 2.4% of the states it occupies (Appendix F). A bit restored to a rail passes at none. No single bit can pass when the noise of the step that reads it includes one amplitude of the change: asked in both directions, its response under Gaussian noise is at most $\Phi(1) - 1/2 = 0.34$, the *binary ceiling*, below the fluctuation.

Rarely at an instant, often within a window. A stationary membrane enters a passing state within 400 ms, the integration time that precedes a conscious percept, with probability 0.085 to 0.997, even when the region it depends on supplies only 1/16 of its noise variance. Some of 10 independent carriers then links two regions within such a window with probability at least 0.59, while at a single instant even 1000 carriers link them with probability as low as 0.025. That is why P is stated per window. The verdict does not rest on the window’s length. No content changes faster than the 20 ms at which two events are perceived in order (Hirsh & Sherrick, 1961), and some of 1000 independent carriers links two regions more often than not from a window of 3 ms upward, and at 20 ms with probability at least 0.98.

Recorded membranes. The same protocol applied to whole-cell recordings from 86 excitatory cells of the barrel cortex of awake mice in quiet wakefulness (Kiritani et al., 2023) tests the declared ranges (Appendix F). The median cell sits 3.7 noise amplitudes below its threshold, farther than the declared 2.0 to 3.5, and passes at 0.15% of its states, inside the declared

range. At a noise share of $1/16$ it enters a passing state within 400 ms with probability 0.15, and some of 10 independent carriers drawn from these cells links two regions within that window with probability at least 0.91.

Readers, shared input and switching nodes. Three variations leave the verdict where it is (Appendix F). A reader that registers only whether the membrane fired within the window finds it passing at 0.013 to 2.4% of its states; if every cortical reader pooled spikes only over windows longer than a content’s time scale, P would deny unity to cortex, and that would count against P. Carriers that share input (Poulet & Petersen, 2008), up to 0.75 of their noise variance, still link two regions: some of 1000 does so within 400 ms with probability at least 0.98. And a chip’s nodes, which pass through their thresholds many times in every window, pass at the voltage as a membrane does, since a change of one noise amplitude moves a crossing by one jitter. The voltage is not what the chip reads, though. Timing closure keeps every transition outside the setup-and-hold window of the latch that reads the node (Ginosar, 2011), so the latch reads the restored bit, which passes at no state and in no window. The separation holds at every yardstick from 0.01 to 0.6 (Figure 2). What separates the membrane from the restored bit is the distance from threshold at which each is held when it is read: a few noise amplitudes, against 506.

Categories. Categorical contents, such as the dominant percept in binocular rivalry, have margins in every substrate, so P is stated on the graded variables from which categories are read, such as the relative strength of competing percepts. This opens no route for a digital machine: every variable it computes is a function of its bits.

Gradual replacement. Replacing a brain’s neurons one at a time by digital units that compute the same input–output function is the standard objection to a substrate condition (Chalmers, 1995). Exact replacement is

unavailable at the grain P concerns: a unit read through margins discards the graded dependence below them, and whether that changes behaviour is what the bridge of Section 6 tests for one connection. Where behaviour and report are preserved exactly, P accepts that unity may still change, because reports are contents and unity is not one more content (Appendix E).

What the premise rests on. Cortex passes the count within a window of a percept’s time scale, which is what P asks, so P rests on wholeness and counting, and every step between them is proved. Counting has a second support in the effective description physics already uses at the parts’ resolution (Appendix D), and a precedent in how it counts clusters in a gas. The bridge tests the two premises together and decides between wholeness and its main rival, joint access, on which contents are unified when available together to the same consumers (Bayne, 2010). The quantized loop of the bridge carries at least as much information to the same regions as the analog loop, in total and about the task. Joint access therefore predicts no advantage for the analog loop, while wholeness, through P, predicts that unity follows the analog loop. Section 8 states what stands if either premise is rejected.

4 When and how far a coupling enforces agreement

P asks two things of the coupling: it must pull the disagreement of unified contents back before the content changes, and it must reach every pair of regions whose contents are unified. The first gives the prediction that the restoration test of Section 6 checks; the second says over what distances that test should find restoration.

Agreement in time. A correction that arrives after the content has changed enforces agreement on a content already replaced, so enforced agreement has

a deadline. Two things limit how soon it arrives. It arrives no sooner than the coupling’s causal cone allows: a site can track a variable at another only after enough rounds of the coupling to place the second site in its causal past. And it arrives at the coupling’s relaxation rate: a coupling that shrinks the overlap discrepancy by a factor $\kappa < 1$ per step brings a discrepancy of d_0 within ε after $\log(d_0/\varepsilon)/\log(1/\kappa)$ steps, and one that shrinks it by no more than ρ per step needs at least $\log(d_0/\varepsilon)/\log(1/\rho)$. In cortex, signal travel typically takes 18 to 31 ms of the 400 ms that precede a conscious percept, so the deadline falls on the relaxation time (Appendix B). P therefore predicts a cortical relaxation time of at most 160 to 166 ms across the cortex, as signals take the slower or the faster fibres, and 171 ms within a posterior zone. A restoration slower than these counts against P.

Reach. The regions to be joined lie up to 15 cm apart if a scene’s contents span the cortex, and about 5 cm apart if they are confined to a posterior zone. On a sheet of coupled oscillators sharing one natural frequency, in the harmonic approximation, the variance of the phase difference between two sites is the noise strength times their *effective resistance*, the resistance a meter would read if each coupling were a wire of the coupling’s conductance. Under coupling of bounded range the resistance grows at least logarithmically with distance, the Gaussian form of the absence of true long-range order in two dimensions (Mermin & Wagner, 1966), while a kernel whose tail falls off more slowly than r^{-4} bounds it at every distance (Chung & Fuchs, 1951; Kunz & Pfister, 1976). A Lipschitz encoder carries phase agreement to content agreement: at noise strength D the root-mean-square content discrepancy is at most $L\sqrt{D R_{\text{eff}}}$ (Appendix A). In cortex the long-range coupling is synaptic, since the field of a local source falls below detection within millimetres (Appendix I).

Local unity and switching. P asks for agreement only among the regions whose contents are unified, which may form one patch of cortex or several

patches joined through a hub that routes activity between them. A hub that routes discrete messages, passing thresholded signals, exchanges records, and records drop out of every chain of couplings. Patches joined only through it are several unified islands whose agreement with one another is computed, as on the switch model of the split brain (Bayne, 2010). A hub that reads the patches' graded state and drives them gradedly is a graded reader and joins them into one system, even if it visits them in turn, since P is stated per window. When the hub is graded, the regress stops at the hub, which is more coupling and not a reader of records.

The thalamus is the natural candidate hub. Its relay cells fire in two modes (Sherman, 2001). In the tonic mode the output follows the input nearly linearly (Lesica & Stanley, 2004). In the burst mode the cell answers with an all-or-none burst, typically when excitation follows a prolonged inhibition (Lesica & Stanley, 2004), and it can burst again only after a renewed hyperpolarization (Sherman, 2001). The protocol of Section 3, applied to a model relay cell with both modes (Smith et al., 2000) at matched rates, orders them: within 400 ms a tonic cell enters a passing state at least 2.3 times as often as a bursting one, in every declared regime (Appendix F). P therefore predicts that unity across patches joined through the thalamus is greater in the tonic mode, a prediction that relay cells recorded in each mode test. The same count applies to machines: analog patches joined by a digital router carry several unified contents and no single one. Bridging two areas within one patch tests P directly, and bridging two patches across a putative hub tests whether the hub is graded.

5 Current artificial systems

A clocked digital chip is a collection of separate records. Each bit is held in place by its own noise margin, and below that margin it varies independently of every other bit and of the voltages beneath it. The records agree only when

the program, applied at each clock step, makes them agree. A transformer running on such hardware reads every content from bits held within their margins, so its agreement is computed, whatever it computes. A chip tracking the bird holds each area’s estimate as a word of bits, and the two words agree when a step of the program copies or averages one into the other; between steps, neither moves with the voltages beneath the other. Hardware whose contents are graded physical quantities coupled without barriers, such as analog, in-memory or neuromorphic substrates, can satisfy P. Satisfying it means that such a substrate’s contents, the column currents of an analog crossbar for instance, are held together physically; it does not determine whether any of them is experienced, or whether any is like a human percept.

Symbols and physical contents. The line P draws is between symbolic and physical content, and the line between digital and analog hardware is its most common instance. A *symbol*, in the sense the margin lemma makes precise, is a content read through a noise margin: a physical state whose identity survives every small enough change of the state that carries it, so two symbols agree only when a rule makes them agree. A *physical content* is a graded quantity that the system’s own dynamics reads, so its agreement with the contents it overlaps is produced by the coupling between them. A bit and a spike count latched at a clock edge are symbols; a spike time, a membrane potential and the phase of a coupled oscillator are physical contents.

Organisation, not material. The criterion concerns how a substrate is organised, not what it is made of. An analog crossbar whose column currents settle through shared conductances, or an array of coupled electronic or optical oscillators, couples its contents as a cortical sheet does. A neuromorphic chip passes where its contents are read from membrane potentials coupled continuously, and fails where they are read from spike counts latched at a clock edge, and a neural circuit read only from all-or-none events latched far from threshold would fail as a digital circuit does. In analog in-memory ac-

celerators, a layer's products are currents summed on the lines of a memory array and then digitized: in a 64-core phase-change chip each row's current is converted into integers, and the traffic between cores runs on those integers (Le Gallo et al., 2023). Such a chip fails P between layers, since its graded stage lies inside each product and couples no two contents.

Why spikes are not bits. What makes all-or-none spikes graded carriers is their timing. When a membrane rises through threshold, the moment it crosses moves continuously with any small input, and receiving cells integrate spike times as they arrive. A clock alone does not separate the substrates: a spike time read through a clock still passes near threshold. What a digital circuit adds is restoration. Each gate drives its output back to a rail, so every transition a latch reads starts and ends far from any threshold. Clockless logic fails in the same way: its transition times move with every voltage, but its values are read from completed codes held at rails (Appendix G).

Leakage and the margin's width. Leakage below a bit's margin, from supply noise, crosstalk or temperature, reaches content only through the error rate. A gate's restoring gain pulls each bit toward its own rail, and pulls no two bits toward agreement. Nor is the width of the margin a free design parameter. A machine whose contents reached the fluctuation at the noise scale would err at every step, so the line P draws coincides with the line between digital and analog operation (Appendix G).

Attractor basins. Cortex has margins of its own. An attractor network's basin is a dynamical margin: which attractor the network settles in survives every small enough change. The basin gives the attractor's label a margin, but it leaves graded the state that carries the label: the position along a continuous attractor, such as a bump of activity that encodes a direction, and the membrane potentials and spike times within the basin, which downstream cells integrate. A cortical circuit whose contents were read only as attractor

labels, far from the basins' boundaries, would fall under the margin lemma as a chip does, and the weak nudge of Section 6 tests which case cortex is in.

6 Tests

What counts as evidence about experience. Every test below measures decoded agreement, the agreement of the contents available to readout. If a split in readout could always be dismissed as a split in access alone, no observation would bear on P. So the evidence about experience is fixed in advance: responses that combine stimuli across the regions in question when the task requires it, and reports of a single field of experience spanning them. A dissociation counts against P if such responses and reports split while graded coupling between the regions is intact, or stay integrated after it is removed. This standard binds P: in a bridged cortex each region still reads its own coupled carriers, so behaviour is evidence about unity, and a split of unity behind unchanged reports is no answer P can give to a null result.

Separating coupling from information. In every manipulation of cortex available so far, graded coupling and information move together, and stimulating the regions from outside does not separate them. Coherence between cortical regions arises largely from communication between them (Schneider et al., 2021), and where in-phase transcranial stimulation of two regions raised their synchrony, it raised directed information flow with it (Reinhart & Nguyen, 2019). The coupling has to be supplied instead.

The bridge. Interrupt the natural connection between two regions, and reconnect them by stimulating each in proportion to the activity recorded at the other (Figure 1). The reconnection runs through one of two loops.

- The *analog loop* is an analog filter and amplifier driving the stimulator, analog end to end, so its output moves with every graded change in its

input.

- The *quantized loop* is the same filter and amplifier with a quantizer between them, so its output is read from bits and has a margin.

The two loops share one filter shape. They differ in the quantizer and in one cutoff frequency, set below.

Matching the information. The analog loop must carry no more information than the quantized one, and the two must fall on opposite sides of P. Adding noise to the analog loop to match them would make it fail P too, so its bandwidth is lowered instead, keeping the cutoff high enough that the loop settles in time (Appendix H). Transfer entropy counts everything one region’s activity tells about the other’s, and the loops could match it and still differ in how much of the task gets through. So the task variable is also decoded from the activity each loop evokes in the region it drives, and the loops count as matched when the analog loop carries no more information than the quantized one, in total or about the task. Both loops were simulated on a model pair of regions, with a linear and a nearest-neighbour estimator of each quantity. At a step of 10 noise amplitudes, a cutoff between 17 and 95 Hz matches the linear estimate of transfer entropy in every simulated regime whose task variable spans at least half a step. At that cutoff the analog loop carries less task information than the quantized one by either estimator, and less transfer entropy by the nearest-neighbour one. An account on which unity follows the information regions exchange therefore predicts no advantage for the analog loop.

The readout. If unity, with the evidence fixed above, follows the analog loop, that is evidence that unity tracks the finer grain and not only the information the regions exchange. If unity is the same under both loops, that counts against P. In animals, unity is measured by integration across the

bridged connection in behaviour and by decoded agreement in the recordings; reported unity needs a bridged human preparation. The loops share every stage but the quantizer and the cutoff, so no property of the electronics separates them, and the bridge tests P as a criterion of organisation (Section 5) against accounts on which unity follows the information regions exchange.

A first preparation. Closed loops that trigger stimulation at one cortical site from spikes recorded at another already run for days in behaving animals (Jackson et al., 2006; Guggenmos et al., 2013); the analog arm replaces the trigger with a continuous drive. A first preparation places chronic arrays in two areas that represent a shared variable, such as the visual and parietal areas that represent the bird’s location, interrupts the natural connection between them, and alternates the two loops across blocks in the same animal. Transfer entropy (Schreiber, 2000) is estimated from each block’s recordings, and the step and cutoff are set on pilot blocks. The primary readout is behavioural: a task that requires combining the values the two areas represent, such as judging whether stimuli in their two fields match. Blocks with the connection interrupted and neither loop running come first, as a positive control: the comparison is informative only if the analog loop restores some of the integration that interruption removes. The companion’s excess decoded agreement is the neural readout. At declared effect sizes the behavioural comparison is the weaker and needs 1248 trials per loop. The loops record and stimulate in separate frequency bands, so the recordings stay readable under a continuous drive (Appendix H).

A weak nudge should spread. Weak stimulation of one region, below the level that changes behaviour, should shift the content decoded in an overlapping region toward agreement with it, in proportion to the stimulation’s strength. Transcranial alternating current and weak intracortical fields supply the nudge. The decoded content must be a graded variable read from

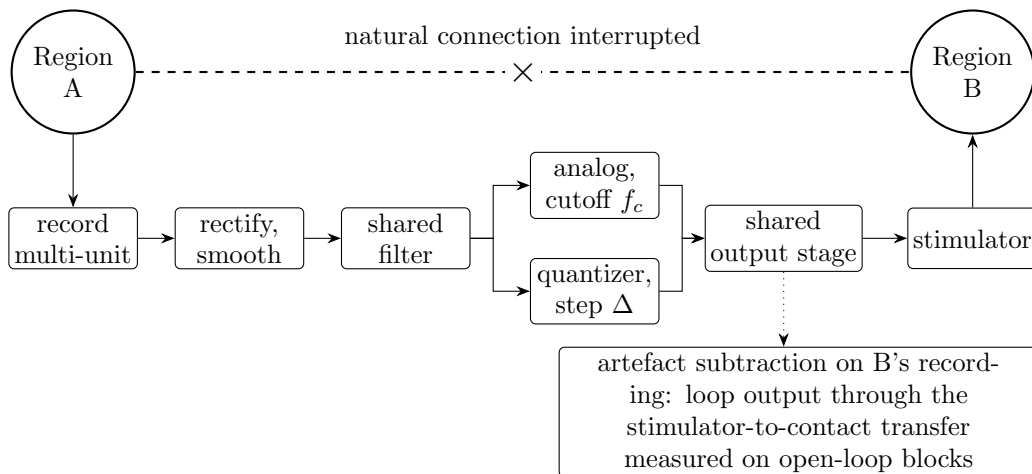


Figure 1: The bridge, one direction of two. Multi-unit activity recorded in region A is rectified and smoothed, passed through a filter both loops share, and then either kept analog or quantized, before an output stage both loops share drives the stimulator in region B. The quantizer’s step and the analog loop’s cutoff are set so that the quantized loop carries at least as much information as the analog loop, in total and about the task. What leaks from the stimulator into a recording contact is removed by subtracting the loop’s own output, passed through the measured transfer. The loop from B to A is the same.

activity the region’s targets integrate, since a decoded category is constant away from its boundaries in any substrate. A change alone confirms little, since most accounts expect stimulating one region to alter connected ones. What P predicts is the change’s form: a shift with the sign that agreement requires, in proportion to strength down to the weakest strength that measurably moves the stimulated region, with no strength below which it vanishes. No shift, a shift of the wrong sign, or one that appears only above a threshold strength counts against P.

Agreement should be restored. Perturb one region and measure how fast its disagreement with its partners comes back down. P predicts restoration within the relaxation time of Section 4, with a restoring pull proportional

to the perturbation, below any threshold as well as above. On an artificial system, patching the activations of one layer, head or position runs the same protocol, and the margin lemma fixes the outcome: restoration by computation above the scale at which the logical trajectory changes, and nothing below it. That contrast makes the cortical outcome informative.

The field’s causal cone. The brain’s electric field reaches the tissue with no propagation delay (Gratiy et al., 2017), while synaptic influence across the cortex takes at least 8 ms. Re-agreement faster than the synaptic route allows would reveal a non-synaptic part of the coupling. The field of a local source falls below detection within millimetres, so the cone is tested at millimetre distances, and at centimetres P predicts re-agreement through synaptic coupling, which is graded as well (Appendix I).

Split brain. Callosal section, read as divided perception with undivided consciousness (Pinto et al., 2017), contested (de Haan et al., 2020), or as consciousness alternating between the hemispheres (Bayne, 2010), removes graded coupling and information together, and the remaining subcortical routes carry both at once: the confound the bridge is built to break.

7 Related work

Appendix J relates P to accounts that tie consciousness to biology, to the analog–digital distinction and to field theories.

Indicator properties. Butlin et al. (2023) assess current AI systems against indicator properties of consciousness, each stated computationally, so every indicator sits at the grain of an algorithm. P adds one indicator stated at the grain of graded dynamics, a condition on how the substrate realises the others, so a system can have every computational indicator and fail P, as a faithful simulation of cortex does.

Computational closure. On the account of Rosas et al. (2024), a macroscopic level that is informationally, causally and computationally closed marks emergent software in a physical system. That closure is the property Section 3 finds at the content level of a digital system, and P reads it the other way round: a level closed off from the graded state beneath it is one whose agreement is computed, not enforced.

Designed to suppress its physics. Kleiner & Ludwig (2024) argue that if a system’s physical dynamics is relevant to its consciousness, processors designed to suppress deviations from their computational dynamics cannot be conscious. The margin lemma states that suppression formally, and this paper adds a scale fixed by the parts’ own noise, which counts systems instead of asking whether physics is relevant at all, a contraction condition, and a test in cortex.

Integrated information. Integrated information theory also denies experience to digital simulations of a brain (Tononi & Koch, 2015; Findlay et al., 2024), through the causal structure of the hardware at the grain of its logic gates. P turns instead on how content depends on graded physical state, measured by one protocol on a cortical membrane and on a logic node.

8 Limitations

- *The premise.* P is a premise and is not derived from the dynamics; the cortical tests bear on whether it holds. It is necessary for unity and not sufficient for consciousness: coupled pendulums satisfy it. It constrains how a system’s conscious contents are held together and is silent on which contents are conscious and what they are like. An analog cross-bar that passes P holds its column currents together physically, which makes none of them an experience, still less one like a human percept.

- *Wholeness.* An account on which unity is joint access to contents rejects the argument's first step. Rejecting wholeness leaves the theorems, the count and the measurements standing, and removes the step from unity to one physical system; the bridge then tests joint access against wholeness directly.
- *Counting.* Rejecting counting, in favour of a count at no particular scale, makes every chip one system. Chips still fail P by the noisy margin theorem; what is lost is the argument from wholeness to P. The count is stricter than a product decomposition: a carrier that registers some changes between its noise and its threshold and not others is two systems by the count and may be one by a product decomposition. Section 3 reads such a carrier as registering through a margin or a threshold, which holds where the response grows with the change. The formal results start from the count.
- *Functionalism.* A functionalism that states its roles at the grain of graded physical state agrees with P on every test here, and it is the intended reading of P. The tests ask which grain unity tracks.
- *The hardware counts.* The ceiling on one bit holds for symmetric single-peaked noise laws that put less than twice the fluctuation of their mass within one standard deviation of the centre. Thermal noise meets it; a bit under a Student law with three degrees of freedom can pass. The thermal and switching counts declare their capacitance, clock, activity, transition time and synchronizer.
- *The cortical estimates.* The single-input estimate combines in vitro measurements of one connection type and one noise recording, through a linear threshold crossing. The occupancy computation models one integrate-and-fire membrane under white noise, with a threshold without dynamics of its own, and its time constants, rates, windows, noise

shares and carrier counts are declared ranges, not fits. The recorded check covers one area of one species in one dataset without simultaneous recordings, and reads its passing states off a Gaussian model held at the slow part of the potential, which predicts 0.47 times the median cell's firing rate. Correlation is one shared component at declared shares, compared with the correlations of neighbouring cells, and omits the unshared input that drives the single spikes of awake cortex (Poulet & Petersen, 2008). The carrier counts stay far below the anatomical ones. The per-window bound is proved for independent carriers and, without independence, as the variance of the number that pass over the square of its mean, a ratio that correlation keeps above a floor. The sweep of the window's length assumes independent carriers, and shared input lowers the link most in the shortest windows. The relay cell is a reduced model with no reticular feedback.

- *Reach.* Phase variance equals effective resistance in the harmonic approximation at identical natural frequencies. The logarithmic lower bound is proved for kernels of bounded range, and the bound under a tail slower than r^{-4} for a sheet that supplies the site counts and couplings of a dyadic chain of discs. Unbounded resistance under a kernel of finite second moment is the classical recurrence criterion on the lattice; logarithmic growth is established numerically. Above the order-disorder crossover the harmonic identity fails. One oscillator population per content patch is a modelling choice about cortex (Appendix A).
- *Which variables are contents.* Which variables a system's own dynamics reads is settled system by system from its circuitry, and the margin lemma takes it as given. A cortex whose unified contents were attractor labels alone would fail P.
- *Artificial systems.* Confirming the predictions for artificial systems

checks that hardware runs inside its margins and leaves P untested; the tests of P are the cortical ones.

- *Agreement in time.* Its edges are proved for a given contraction factor, and identifying that factor with the coupling's spectral gap is linear response. The relaxation time of cortical coupling is unmeasured, and agreement is read as a declared reduction factor of disagreement.
- *The field.* A long-range tail improves agreement only in proportion to its conductance against the short-range part. The field's reach rests on an estimate that places its peak at a declared distance, and its contribution across centimetres is a quantity to be measured.
- *The bridge's reach.* A functional account that counts the quantizer's nonlinearity, or the cutoff that matches information, as a difference between the loops can accommodate either outcome; such an account individuates roles finely enough to see a quantizer, which is the grain P names, so the bridge leaves it on P's side of the line. The comparison presupposes that the analog loop restores some integration, which the positive-control blocks check. That the loops can be matched is shown on a linear model pair of regions.
- *What a null leaves open.* The bridge routes coupling through electronics and stimulation, so it tests P as a criterion of organisation. A null, in which both loops restore integration equally, counts against that reading and is compatible with a substrate-bound variant on which graded coupling must run through tissue, a variant no artificial bridge can test. Both arms also share two auxiliaries: stimulation enters a region unlike synaptic input, recruiting cells synchronously and antidromically, and the loop realises P only at the grain of a multi-unit envelope recorded from a few contacts.
- *Human evidence.* No human preparation yet measures reported unity

across a bridged connection. Human data can falsify P but cannot by themselves tell the graded grain from the grain of information exchanged, because the two co-vary in every manipulation available so far.

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A Reach: resistance, encoders and simulations

The logarithmic growth is a lower bound for every kernel of bounded range. Group the sites into shells around a site a , so that the coupling joins only sites in the same or adjacent shells. The shells then act as conductors in series, and the effective conductance from a to any site beyond shell n is at most $(\sum_{k < n} 1/C_k)^{-1}$, where C_k is the coupling crossing from shell k to shell $k+1$. On a planar sheet the perimeter of shell k grows linearly in k , so $C_k \leq c(k+1)$ for a constant c set by the kernel’s strength and width, and the effective resistance is at least $\log(n+1)/c$. Widening the kernel enlarges its shells and raises c , which slows the growth and leaves it unbounded. Distance-independent agreement on a sheet therefore needs coupling of unbounded range.

A tail slower than r^{-4} bounds the resistance at every distance. Run a chain of site sets out from a through discs of radius 2^j and back in to b , the two widest discs spanning the separation. The phase difference across a and b telescopes through the discs’ mean phases, and the drop between the means of two consecutive discs is bounded by the coupling energy between them, shared among the pairs of their sites. A disc of radius 2^j on a planar

sheet holds at least $\alpha 4^j$ sites, and a kernel $K(r) \gtrsim r^{-(2+\sigma)}$ couples two sites of consecutive discs by at least $\beta 2^{-(2+\sigma)j}$. The number of pairs grows as 16^j and their coupling falls as $2^{-(2+\sigma)j}$, so the drops shrink geometrically with ratio $q = 2^{-(2-\sigma)/2}$, which is below one for $\sigma < 2$, and the effective conductance is at least $(1 - q)^2 \alpha^2 \beta / 8$ however far apart a and b lie.

Both results match classical potential theory. Imagine a random walker on the lattice that jumps between sites with probabilities proportional to the kernel, which is the same at every site. The resistance from a site to infinity is infinite exactly when that walker is sure to return to its start, that is, when the walk is recurrent (Doyle & Snell, 1984). Shorting every site outside a disc around a can only lower the resistance from a to a site b beyond the disc, so under recurrence that resistance grows without bound as b recedes. A symmetric kernel of finite second moment, which includes every tail steeper than r^{-4} , gives a recurrent walk in two dimensions (Chung & Fuchs, 1951). The bound above is the matching statement for tails slower than r^{-4} , whose walks escape. The same threshold holds beyond the harmonic approximation: the plane rotator in two dimensions with coupling falling as $r^{-\alpha}$, $2 < \alpha < 4$, has true long-range order at low noise (Kunz & Pfister, 1976), which short-range coupling forbids. The plane-rotator sheet also has a Kosterlitz–Thouless transition (Fröhlich & Spencer, 1981), an order–disorder crossover that the simulations below locate. When natural frequencies differ from site to site, short-range coupling fares worse still: phases then desynchronize in every dimension up to four (Hong et al., 2005).

Two further properties bound the resistance from above. Adding coupling never raises it, and a direct coupling caps it at its reciprocal. Across a finite cortex a wide short-range kernel and a slow tail differ by a slowly growing logarithm, so telling them apart takes a measurement of the kernel’s second moment as well as of its tail.

An encoder that reads the phases of a small neighbourhood relative to the phase at its centre does better than the Lipschitz bound of Section 4.

It reports one content at every site of a travelling plane wave or a winding, whatever the gradient, and for a wave plus a departure from it its discrepancy between two sites is at most its Lipschitz constant times the departure's relative readings there, so it pays for the local roughness of the field and for nothing else. Both bounds read one oscillator population per content patch. Across centimetres, where two territories describing one quantity need not be neighbours on the sheet, that reading is an assumption, the *phase-to-content assumption*: each territory's content is read, by encoders of Lipschitz constant L that are equal where territories overlap, from the phase of one oscillator population assigned to it, so that the phase agreement of the two populations bounds the agreement of the two contents. It is a modelling choice about cortex, since the coupling of a field is spatial while two territories describing one quantity need not be neighbours. Under it the coupling's reach bounds phase disagreement between distant territories directly and content disagreement through it.

Simulated sheets at identical frequencies, of side 64 and 128, with nearest-neighbour, exponential and power-law kernels of tail exponent $\sigma = 1$ and $\sigma = 3$ at a common row sum, follow the harmonic identity in the ordered phase. The measured phase variance is at most 1.15 times $D R_{\text{eff}}$ at $D = 0.1$ and at most 1.26 times at $D = 0.4$, the excess rising with noise as the cosine coupling softens, and a logarithm fits its growth with distance better than a line in 16 of the 16 runs at those noise levels. At $D = 1.2$ the nearest-neighbour sheet disorders: the mean chord discrepancy $\langle 1 - \cos(\theta_a - \theta_b) \rangle$ at the largest separation is 1.00. The exponential and $\sigma = 1$ sheets stay ordered at that noise (0.46 and 0.50), and the $\sigma = 3$ sheet nearly disorders (0.92), so the kernel's width sets the order-disorder crossover as well as the rate of decay. Exact resistance sums at sheet sides 128, 512 and 1024 separate width from tail. The exponential kernel, with 25 times the nearest-neighbour stiffness, gives resistance at half the side of 0.531, 0.549 and 0.558, still growing; the $\sigma = 1$ tail gives 0.580, 0.577 and 0.577, and the $\sigma = 3$ tail, steeper than r^{-4} ,

gives 1.198, 1.448 and 1.572, growing as a short-range kernel does.

B Agreement in time for cortex

The causal cone is stated precisely as follows: a report at one site that tracks a variable at another to within ε , for two values more than 2ε apart, is possible after T rounds of the coupling only if the second site lies in the first site’s causal past for T rounds.

The longest cortico-cortical fibres run about 15 cm (Nunez & Srinivasan, 2014). Callosal axons conduct at median velocities of 4.9 to 8.8 m/s, and the fastest of them at up to 20 m/s (Caminiti et al., 2009). A chemical synapse adds about 0.5 ms (Katz & Miledi, 1965; Sabatini & Regehr, 1996). The earliest a synaptic signal can cross the cortex is therefore 8 ms, along the fastest axons, and typical arrival takes 18 to 31 ms. For the time scale of a content, take the unconscious integration that precedes a conscious percept, which lasts up to 400 ms (Herzog et al., 2016). Count agreement as reached when disagreement has fallen 10-fold, decaying exponentially. Agreement then arrives in time when the coupling’s relaxation time is at most 160 ms with typical arrival along the slower fibres, and 166 ms along the faster. Over a localized extent of 5 cm the earliest arrival is 3 ms, typical arrival along the faster fibres is 6.2 ms, and the relaxation time may reach 171 ms. Signal travel thus uses only a small part of the time available, and whether agreement arrives in time depends on how fast the coupling settles.

C A single input and noise that changes with the state

A neuron in numbers. Suppose an input ΔV , too small to fire the cell on its own, arrives while the membrane potential is rising through threshold at rate $\dot{V}$. It moves the spike by $\Delta V/\dot{V}$. Voltage noise σ_V jitters the spike by

$\sigma_V/\dot{V}$. The shift therefore exceeds the jitter exactly when ΔV exceeds σ_V , whatever the slope, and spike generation adds little noise of its own (Mainen & Sejnowski, 1995). A unitary excitatory potential between cortical pyramidal cells averages 1.3 mV, with a coefficient of variation of 0.52 (Markram et al., 1997), and the membrane noise near threshold is 0.54 mV (Jacobson et al., 2005). Counting the input’s own variability as noise, one such input shifts the spike by 1.5 times its jitter, a response of 0.55. Every input larger than 0.64 mV passes, up to the input that fires the cell.

Noise whose law changes with the state. When the law of the noise changes with the state, as the variance of a spike count does, a larger change still moves the content at least as much as a smaller one on each side of the state, provided the laws have a monotone likelihood ratio, as exponential families do: such laws cross once along the parameter the region moves. The criterion of Section 3 is then met exactly when a change of one noise amplitude up and one down both reach the fluctuation.

D Counting in detail

The logic node. A change of one amplitude toward the threshold moves a restored bit by the fluctuation only from half an amplitude away, 0.099% of the margin’s width, where the bit is in the middle of switching, and a change in both directions moves it by less at every distance, by the binary ceiling. No other bit registers a change anywhere in the margin except through the error rate. By the noisy margin theorem, once that rate is under the fluctuation and each carrier has a change above its noise inside the margin, no carrier of any region registers any other, and the per-window form of P fails exactly as the per-instant form does.

The chip’s wires pass on only changes that cross a threshold, which is the rule applied at each clock step. Crosstalk and supply noise below the

margin are real interactions, and they move the voltages. By the noisy margin theorem they move the contents by at most the error rate, the quantity a working chip is designed to keep small. The count looks strange only because ordinary usage counts by program, and one program uses all of the bits. Counted by physics, each bit is restored to its own rail by its own gates, and what keeps the bits in step is the rule the program applies, a protocol among separate records. The citizens of Block’s nation likewise register each other only through the messages they exchange.

The same count from effective descriptions. Physics reaches the same count a second way, through the effective description it uses at the parts’ own resolution. Such a description keeps the variables that survive coarse-graining between the parts’ noise and their thresholds, and it treats as separate degrees of freedom exactly those whose dynamics it can write without reference to one another’s state. A chip has one such description, its logic. Each bit evolves, at each clock step, by a Boolean function of the bits wired to it, and the voltages beneath are integrated out at the cost of the error rate. This is the closure of the content level defined above, the property that marks a macroscopic level as computationally closed (Rosas et al., 2024), and in it the bits interact only through the rule. A coupled field has no description of that kind at the same resolution. Averaging a sheet of coupled oscillators or membranes over blocks leaves a sheet whose blocks pull on their neighbours, and each block’s dynamics is written with its neighbours’ state in it. So the effective description makes a chip’s contents many systems and a coupled field’s contents one, as the noise-floor count does. It is the description an engineer uses to design the chip and a physicist uses to model the field, not one built to separate them.

E Gradual replacement

The standard objection to a substrate condition replaces a brain’s neurons one at a time by digital units that compute the same input–output function (Chalmers, 1995). If behaviour and report are preserved throughout while unity fades or divides, then at the intermediate stages the subject is wrong about its own experience, which the argument takes to be absurd. P answers in two parts.

First, exact replacement is unavailable at the grain P concerns. By the margin lemma, a unit that reads its inputs through margins reproduces a neuron’s function to the resolution of its margins, and discards the graded dependence below it. Whether discarding it changes behaviour is an empirical question. The bridge of Section 6 asks it for one connection, so the bridge is a local, testable instance of replacement.

Second, where replacement preserves behaviour and report exactly, P accepts the conclusion the argument calls absurd, and gives its reason. Reports are contents computed from contents, and unity is not one more content, so reports can stay the same while unity changes. Behaviour is evidence about unity in a system whose contents are carried by its own coupling, and a system built to reproduce that evidence by computation is the simulation objection again. A bridged cortex is of the first kind, so at the bridge P stakes itself on behaviour (Section 6).

F The occupancy computation

What the membrane is asked. The membrane’s next content is the time of its first spike within 10 or 30 ms, and every change of one to 3 noise amplitudes, in either direction, must move that content by the fluctuation. Its time constants span the in vivo high-conductance state and its rates the sparse rates of awake cortex. The passing fractions of Section 3 are for a region that supplies all of the membrane’s noise. A bit restored to a

rail at the logic node's margin passes at none of its states, and its largest response over them is below 10^{-100} . The binary ceiling stays below the fluctuation under every symmetric noise law with one peak that puts less than twice the fluctuation of its mass within one standard deviation of its centre, logistic and Laplace noise included, as shown at the end of this appendix. A graded content clears the fluctuation by a wide margin: at its best state the membrane responds by at least 0.64 in every regime and at every noise share.

Windows. For independent carriers the chance that none passes within a window falls geometrically with their number: with 100 carriers some carrier passes within 40 ms with probability at least 0.59. Reading the spike time through a clock of 1 or 5 ms leaves the passing fraction unchanged. The value a latch reads links in no window of any length.

Readers that ignore spike times. A reader that registers only whether the membrane fired within the window, and discards every spike time, passes with a next content of 10 ms at the same share of states in every regime as a reader of spike times. At a membrane a few amplitudes below threshold, what a change of one noise amplitude moves is the chance that it fires within the window, and a spike count registers that; a count responds at least as much as this reader, since whether the membrane fired is a function of the count. So P classes a cortical membrane as a graded carrier whether its readers integrate spike times at the grain of milliseconds, as the evidence of Section 2 indicates, or pool spikes over the content's window.

Shared input. In awake cortex neighbouring membrane potentials are highly correlated during quiet wakefulness and less so during active behaviour (Poulet & Petersen, 2008). The model gives every membrane a shared component carrying a fraction c of its noise variance, so that two membranes correlate by c and each keeps its own law. At least one of 100 such carriers

enters a passing state within 400 ms with probability at least 0.88 for every share up to $c = 0.75$, in every regime.

Switching nodes. Take a 1 GHz clock and nodes that switch in 0.02 to 0.5 of its cycles, each transition lasting 20 ps. If a node’s next content were the time of its crossing, the node would pass on the rising half of each transition, at 0.02 to 0.5% of its states, within the membrane’s range, and within every window of 40 ms except with probability below 10^{-100} . By the carrier condition of Section 2, though, a node’s next content is what its latch reads at the clock edge, the node at its rail. The crossing a latch discards is exactly the content a membrane delivers when its readers integrate spike times as they arrive. The separation holds for count readers too: a reader that registered only whether a node switched within a cycle would take a quantity fixed by the node’s logical inputs, which are restored bits, moved by its noise only through the error rate.

A thalamic relay cell. The integrate-and-fire-or-burst model (Smith et al., 2000), with its published parameters, adds to the membrane a low-threshold calcium current that flows above -60 mV, inactivates there and recovers below it. Noise gives the passive membrane a standard deviation of 2 or 4 mV, the unit of every change. In burst mode a constant current holds the membrane below the calcium threshold, where the current recovers and noise that carries the membrane past the threshold sets off a burst; in tonic mode it holds the membrane above, the current stays inactivated, and the cell is an integrate-and-fire membrane. In each mode the current is calibrated to a rate of 0.5 or 2 Hz, and the next content is the first spike time within 10 or 30 ms, as for the cortical membrane, with the calcium current’s state kept under each change. At an instant the two modes pass at similar shares of their states, 0.16 to 1.5% in tonic mode and 0.3 to 1.6% in burst mode. A bursting cell’s passing states lie in and around its bursts, which are rare, while a tonic cell’s lie below its spike threshold, where it spends the

time between spikes. Within 400 ms a tonic cell enters a passing state with probability 0.28 to 0.84, and a bursting cell with probability 0.07 to 0.34. At matched noise, rate and window the tonic cell’s chance is at least 2.3 times the bursting cell’s, within 40 ms as well. Probability is propagated exactly on a grid in membrane potential and inactivation, and a run at half every grid step moves no visit probability by more than 8.5%.

Recorded membranes. The recordings (Kiritani et al., 2023) hold 240 cells of layers 2/3 to 5 of the barrel cortex of awake head-restrained mice, with quiet wakefulness, free whisking and touch scored from whisker filming; the excitatory cells are the membranes the model describes. Spikes are detected by the authors’ criterion for each cell, a cell’s threshold is the median potential at spike onset, and each spike, from onset to repolarisation, counts as a failing state, as the model’s refractory states do. The awake potential moves on two time scales: fluctuations with a correlation time of 11 to 17 ms, and a wander over hundreds of milliseconds that carries much of its variance. One time constant fitted to both misses the fast fluctuations that set the next spike, so the potential is split into its running mean over 100 ms and a fast remainder. The noise amplitude and the model’s time constant are those at which the model membrane, analysed exactly as the recording is, has the remainder’s spread and correlation time. Within a next-content window the membrane is taken to be the model membrane held at the mean behind the running mean’s current value, which supplies the law of the next spike time that a single recorded path does not. Across the 86 cells with at least 10 s of quiet wakefulness, the tenth to ninetieth percentiles run from 2.2 to 5.1 noise amplitudes below threshold, from 0.025 to 1.3% of states passing, with 90% of cells inside the declared range, and from 0.01 to 0.51 for a passing state within 400 ms at a noise share of 1/16; for 31% of cells that chance is below the least declared regime’s, 0.085. During free whisking (73 cells) the median cell sits 4.1 amplitudes below threshold and enters a passing state

within the window with probability 0.13. Splitting at 200 ms instead gives a median passing share of 0.12% and a median visit probability of 0.13. For the median cell the model predicts 0.47 times the recorded firing rate, and it comes within a factor of two for 48% of cells: recorded membranes fire more readily than a Gaussian membrane at the same distance, and the passing states are read off that Gaussian description. None of the 689 sweeps timed to the second holds two cells recorded at once, so the share of input that neighbouring cells hold in common stays declared.

The yardstick. The separation of the membrane from the value a latch reads holds at every yardstick from 0.01 to 0.6 (Figure 2), not only at the fluctuation. Across that range the latched logic node passes at none of its states, and the membrane enters a passing state within 400 ms with probability at least 0.036 at the highest yardstick, and at least as often at every lower one. Below $\Phi(1) - 1/2$ one bit can reach the yardstick, so there a synchronizer’s first stage can pass, at no more than 0.2% of its samples. The formal results take the yardstick as a parameter and hold at every value of it. The fluctuation is the natural choice because it makes *large* mean what it means for the carrier itself: a change that moves the content by as much as the content moves of its own accord.

Method. The membrane is an Ornstein–Uhlenbeck process in units of its own stationary standard deviation, with the threshold at zero. After a spike it is refractory and then reset to its mean, and the mean is calibrated so that the stationary rate equals the declared rate. Probability is propagated on a voltage grid through the exact Ornstein–Uhlenbeck step, and a crossing between two sampled times is counted with the Brownian-bridge probability $\exp(-2xy/s^2)$ for sub-threshold samples x and y and step deviation s . Nothing is sampled. From each grid state, the law of the first spike time in the window, with its absence as one more outcome, is propagated in one pass, and the response to a change is the total-variation distance between two such

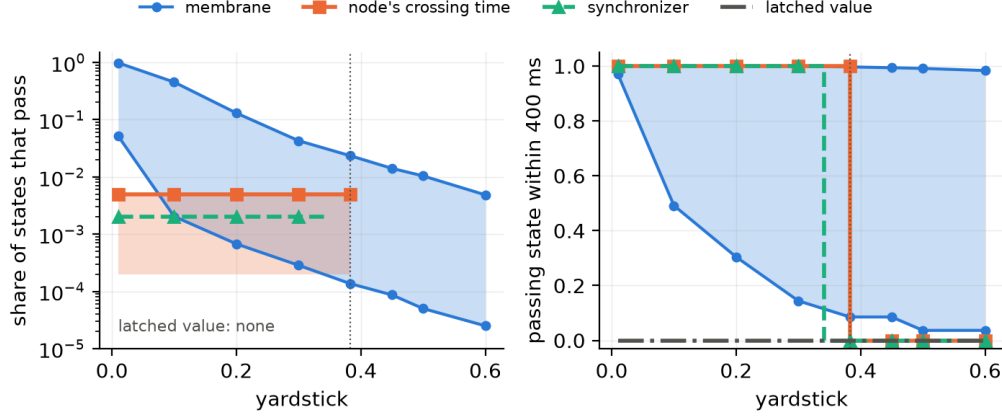


Figure 2: One protocol at every yardstick. Left: the share of occupied states at which every admissible change moves the next content by at least the yardstick. Right: the chance of entering such a state within 400 ms. The membrane is a band over every declared regime and, on the right, every noise share. A switching node's crossing time is a band over the declared activity factors; it passes exactly at the fluctuation (dotted) and at no higher yardstick. The synchronizer's share is the upper bound $T_W f_D$, at the yardsticks one bit can reach, up to $\Phi(1) - 1/2$. The value a clocked latch reads passes at no state and in no window.

laws. A state passes when every admissible change moves the law by at least the fluctuation, where a change is admissible if it is at least one noise amplitude of the carrier and does not carry the membrane across threshold. The fraction of occupied states that pass is the stationary mass of the passing states. The probability of entering a passing state within a window is computed on the same chain with the passing states made absorbing. A carrier that supplies a share φ of the membrane's noise variance has noise amplitude $\sqrt{\varphi}$ in the membrane's units, and its changes are scaled accordingly. The bit is a carrier restored each step to a rail at a declared margin, read after one step of unit Gaussian noise, with occupancy given by that noise around the rail. The same computation is repeated at yardsticks from 0.01 to 0.6 in place of the fluctuation, at every regime and noise share. For correlated car-

riers, each membrane is the sum of a shared Ornstein–Uhlenbeck component of variance c and a private one of variance $1 - c$, both with the membrane’s time constant. Given the shared path the carriers are independent, so the probability that none of K passes within a window is the average of $(1 - h)^K$ over shared paths, where h is one carrier’s chance of passing given the path. For each of 256 shared paths from a fixed seed, the membrane’s probability is propagated through the exact step with the private noise, its centre moved by the shared path’s increment, after a burn-in of five time constants. Halving both the grid spacing and the time step changes the passing fraction by 10%, and the correlation of two carriers’ passing by at most 0.012.

Correlation lowers the per-window link. The chance that 100 carriers leave a window of 400 ms without a link is at most 0.0018 at $c = 0.25$, 0.017 at $c = 0.5$ and 0.12 at $c = 0.75$. These bounds, and the links of at least 0.88 and 0.98 in Section 3, lie two Monte Carlo standard errors beyond the estimate in every regime. The shorter window is more sensitive: within 40 ms the link falls to about 0.23 at $c = 0.75$, so P’s per-window condition is met at the time scale of a percept rather than of its fastest components.

The window is also swept on its own, from one time step to 1000 ms, for independent carriers. The chance that one carrier occupies a passing state at some step of a window is propagated once on the same chain, so every window comes from one pass, and the link of K carriers within it is $1 - (1 - h)^K$. The link reaches one half in every regime and at every noise share from 3 ms with 1000 carriers and from 31 ms with 100, and since h grows with the window it stays above one half at every longer one. Halving the grid spacing and the time step moves both ends earlier, so these are the conservative values.

The switching chip is counted in closed form. On the rising half of a transition the node’s voltage crosses its threshold at a slew $\dot{V}$, a change ΔV moves the crossing by $\Delta V/\dot{V}$ and the node’s noise jitters it by $\sigma/\dot{V}$, so the response is $2\Phi(\Delta V/2\sigma) - 1$ whatever the slew. Every admissible change of at least one noise amplitude then reaches the fluctuation, exactly at one

amplitude, so the node passes at the fluctuation and at no higher yardstick. A node that switches in a fraction α of the cycles of a clock of frequency f , with transitions of duration t , spends $\alpha ft/2$ of its time on the rising half of one, and a window of N cycles holds no transition with probability $(1 - \alpha)^N$. A synchronizer with aperture T_W , clock f_C and data changing at rate f_D samples inside its aperture at the rate $T_W f_C f_D$, so at most a share $T_W f_D$ of its samples do. Each sample is read as one bit, so it passes at no yardstick above $\Phi(1) - 1/2$, and at lower yardsticks that share bounds its passing fraction. A second stage allowed a time S to resolve fails once in $e^{S/\tau}/(T_W f_C f_D)$ on average, with τ the stage's resolution time constant (Ginosar, 2011); the declared values are that tutorial's 28 nm example, $\tau = 10$ ps and $f_D = 100$ MHz, with S one cycle.

The one place a working chip latches a transition is a synchronizer between clock domains. With a 20 ps aperture and data changing at 100 MHz (Ginosar, 2011), its first stage samples inside the aperture at most 0.2% of the time, 8×10^5 times in a window of 400 ms, and which way it resolves moves with the timing of the transition; a second stage given one cycle to settle leaves it unresolved once in more than 10^{29} years. What that second stage takes is the bit the first resolves to. The noise in what the first stage samples includes the transition's own jitter, one amplitude of the change, and it is thermal, so Gaussian, and one bit under Gaussian noise does not reach the fluctuation: at the centre of the aperture, where its two-sided response is largest, a change of one noise amplitude in the transition's timing moves the probability of either resolution by at most $\Phi(1) - 1/2 = 0.34$, and the stage's own noise only lowers that. The synchronizer therefore fails graded dependence at every sample, those inside its aperture included, whichever protocol carries the data across the boundary, and so does a random source that reads its bits from how a metastable stage resolves.

The ceiling depends on the noise law only through its mass near the centre. A bit read as 1 when a quantity plus noise exceeds a threshold

responds to a change a by the noise law's mass on one of the two intervals of width a that meet at the threshold, so its two-sided response is the smaller of the two, at most half the mass of an interval of width $2a$. For a symmetric law with one peak the smaller mass is largest with the threshold at the state, where it is the mass between the centre and one amplitude. One bit therefore stays below the fluctuation at every threshold exactly when the law puts less than twice the fluctuation of its mass within one amplitude of its centre. Scaled to unit standard deviation, Gaussian, logistic and Laplace noise meet this, and a Student law with three degrees of freedom does not.

Several bits held near their thresholds together can read one graded quantity q , as the stages of a time-to-digital converter read one transition time. The word is a function of q plus the quantity's own noise and the stages' noise, and the stages' noise is independent of q . The response of the word to a change of q is therefore at most the response of q itself, $2\Phi(a/2\sigma) - 1$ under Gaussian noise of amplitude σ , by the data-processing inequality for total variation. With two thresholds half an amplitude on either side of the state and no stage noise, the word's three outcomes split the line where the two laws cross, and a change of one amplitude in either direction moves it by exactly the fluctuation. A word whose thresholds straddle the quantity within its noise thus passes exactly where the quantity passes.

G Digital hardware in detail

Clockless logic. Asynchronous circuits have no clock. In a quasi-delay-insensitive design, each stage signals completion to the next through a handshake, and correctness holds under every assignment of gate and wire delays apart from a few forks assumed isochronic (Martin & Nyström, 2006). Their transition times vary gradedly with every voltage, as a spike time does. But the design makes the computed values independent of those times: a value is accepted once its code is complete, whatever delays produced it. The timing

of such a circuit is therefore a trace in the sense of Section 2. It moves with the physics, while the circuit’s own contents are read from completed codes, each held at a rail by restoring gates, so they meet the margin lemma as a clocked circuit’s contents do.

Leakage. Real digital hardware leaks: supply noise, crosstalk and temperature move the analog quantities behind every bit. The margin lemma says where such leakage can reach content. If a content read as a word of thresholded continuous quantities depends gradedly on any region, some bit of the next state sits exactly on its threshold, and correct operation keeps every state off that set. Digital logic does pull perturbations back, but each gate does so alone: its gain pulls a disturbed voltage back toward its own rail. That restores each bit toward its own logical value, pulls no region toward agreement with another, and is what keeps the bits inside their margins. The chip’s records stay separate even in the one place where its physics contracts.

The margin also has a width. If every bit of the next state is at least δ from its threshold, each bit is read from an L_v -Lipschitz quantity and the micro dynamics is L -Lipschitz, then a change to one region smaller than $\delta/(LL_v)$ changes no bit. Leakage below that scale reaches content only through the error rate (Section 3), and leakage above it flips a bit, which is a logical error. Digital design keeps ordinary noise well below that scale: a textbook inverter’s static noise margin is 41% of its supply voltage, against an expected supply variation of 10% (Rabaey et al., 2003), so the expected variation is 0.24 of the margin. The nondeterminism of floating-point reduction order is a choice among logical trajectories and lies outside both results: it changes which content is computed, not how content depends on graded state.

The margin’s width. If P turned on how wide an engineer happened to make a noise margin, that would be a reductio of P: two machines running one program would differ in unity by a design parameter. The width is not free in that way. By the noisy margin theorem a content’s response to a region

is at most its error rate, so a machine whose contents reached the fluctuation at the noise scale would err with probability at least 0.38 at every step. Its bits would fail to hold their values from one step to the next, its program's trajectory would stop being reproducible, and whatever agreement survived would come from the physics beneath the bits and not from the program. The line P draws therefore coincides with the physical line between digital and analog operation, and working systems sit far from it on either side: a logic node's margin is 506 noise amplitudes wide, while a cortical membrane sits on average 2.0 to 3.5 noise amplitudes below its threshold, and some membrane of a region reaches a passing state within a window of a percept's time scale with high probability.

H The bridge in detail

The analog loop must carry no more information than the quantized one, and the two must fall on opposite sides of P. Section 3 fixes what the second condition means: a change of the recorded region by its own noise amplitude must move the analog loop's output by at least the fluctuation, and the quantized loop's output by less. The analog loop passes exactly when its noise, in the region's units, is at most the region's own. The quantized loop fails at every state farther from a quantization edge than the region's noise plus 0.3 times the recording noise, so its step must be many times both. Matching information by adding noise to the analog loop defeats this. At matched information the added noise is 0.24 of the quantizer's step, so an analog loop that still passes caps the step at 4.1 noise amplitudes of the region, and the quantized loop then fails at no more than 52% of its states, taking states as uniform within a step. The analog loop's bandwidth is lowered instead, with no noise added, until the information it carries, estimated from the recorded activity, no longer exceeds the quantized loop's. Lowering the bandwidth slows the loop, and a loop that settles after the

content has changed fails P’s timing condition (Section 4) however gradedly it couples. A first-order filter with cutoff f_c settles with time constant $1/(2\pi f_c)$, and the bridged coupling relaxes no faster than its loop, so the analog loop’s cutoff is kept at or above 1 Hz, whose time constant is at most the shortest relaxation time that brings agreement in time in Appendix B. If matching would take the cutoff lower, the quantizer’s step is coarsened instead, which lowers the quantized loop’s transfer entropy and raises the share of its states that fail. A null result then cannot come from an analog loop too slow for the content’s window. Above that floor the step is free: at a step of 10 noise amplitudes, with recording noise no larger than the region’s own, the quantized loop fails at 74% of its states or more. Stepping the quantized loop’s resolution checks that its transfer entropy rises with the resolution, so that the resolution, and not recording noise, sets the matched value.

Whether the loops can be matched. Two settings need not meet two conditions, so both loops were simulated on a model pair of regions, at a 1 ms step and in units of region A’s noise amplitude. A task variable, a process of standard deviation 2 to 10 and time constant 20 or 100 ms, drives region A on top of A’s own noise. A is recorded with noise of its own amplitude and passed through the shared filter, then through the analog loop’s cutoff or the quantizer, and the shared output stage drives region B, a leaky integrator with noise of its own. Both quantities are estimated twice. The linear estimator takes transfer entropy from A’s recording to B’s by the linear-Gaussian formula, and task decodability as the variance of the task variable that a linear decoder of B’s recording explains on held-out trials. It misses what the quantizer’s nonlinearity passes on, so the nearest-neighbour estimator takes transfer entropy as a conditional mutual information (Kraskov et al., 2004; Frenzel & Pompe, 2007) and decodability as the variance a decoder would explain if the task information in B’s recording, estimated the same way, were Gaussian; on the analog loop the two estimators agree. For each

step and estimator, two cutoffs are read off a grid from the floor above to 1 kHz: the one that matches the quantized loop’s transfer entropy and the one that matches its decodability. Both measures rise with the cutoff, so at the lower of the two the analog loop carries no more of either.

At the step of 10 noise amplitudes, in every regime whose task variable spans at least half a step, the linear transfer entropy is matched between 17 and 95 Hz, well above the floor, and the matched analog loop explains 0.025 to 0.06 less of the task variable’s variance than the quantized loop. The same holds at every simulated step no larger than twice the task variable’s standard deviation. Lowering the bandwidth removes fast activity that carries transfer entropy and little of the task, while the quantizer removes small changes at every frequency, so at matched linear transfer entropy the quantized loop keeps more of the task. When the task variable spans only a fifth of a step, the matched analog loop can carry more task information, by up to 0.075, and the step has to be made finer before the loops are compared. The task is therefore chosen so that its variable spans several noise amplitudes of the recorded region.

By the nearest-neighbour estimator the quantized loop carries more transfer entropy than the analog loop does at any cutoff up to 1 kHz, because its jumps tell the driven region when the recorded one crossed a step, so at the declared step no cutoff matches it. At the steps where one does, the matched analog loop carries more task information in 4 of 7 cases, by up to 0.038: which side of a transfer-entropy match the task information falls on depends on the estimator. The lower match does not. The nearest-neighbour decodability is matched between 33 and 399 Hz, above the linear transfer-entropy match in every regime, so the lower match is the linear one, and a preparation sets the cutoff there. At that cutoff, by the nearest-neighbour estimator, the analog loop carries 10 to 55% less transfer entropy and explains 0.028 to 0.065 less of the task variable’s variance than the quantized loop, and it carries less of both at every simulated step no larger than twice the task

variable’s standard deviation.

On the preparation, the decoder is fitted on other blocks of the same loop, the step and the cutoff are set together on pilot blocks, and the check is repeated on the blocks that are compared.

With blocks of 100 trials repeated on independent noise, the matched transfer entropy and the gap vary from block to block, and their confidence intervals narrow with the square root of the trials. The behavioural comparison, accuracy 0.75 against 0.7 at a two-sided level of 0.05 and power 0.8, needs 1248 trials per loop, and the neural comparison, at a standardized difference of 0.2, needs 393. At the behavioural count, by either estimator, the transfer entropies are known to within 4.4% and the difference in task variance to within 0.006, smaller than every margin above, so the pilot blocks can confirm that the analog loop carries no more of either. The model has no spikes, no stimulation artefact and no plasticity.

The analog loop has to be analog end to end, because a loop that digitized the recording before driving the stimulator would run the quantized condition in both arms.

Stimulation artefact. Event-triggered loops blank the recording around each pulse, and a continuous drive leaves no interval to blank. The loops therefore record and stimulate in separate frequency bands. Each loop reads multi-unit activity, above a few hundred hertz, takes its envelope, and drives the other region with a current that follows the envelope and so lies in the band below it. Rectifying and smoothing are analog operations, so the analog loop stays analog end to end. What leaks across the bands is removed by subtracting the loop’s own output, which the loop knows exactly, passed through the transfer from stimulator to recording contact measured on open-loop calibration blocks. Both loops end in the same output stage, which smooths the quantizer’s steps into the stimulation band, and share the stimulator, the bands and the subtraction, so a residual artefact reaches both arms by one

path. Open-loop blocks in which each stimulator follows activity recorded in another block measure that residual in each arm, and the comparison of the loops is made after it is removed.

I Timing the field’s cone

Brain tissue carries an electric field as well as synapses. At the frequencies of neural activity that field is quasi-static (Gratiy et al., 2017): it reaches every point of the tissue it spans with no propagation delay. Synaptic influence between distant regions, by contrast, arrives only after axonal conduction and synaptic transmission, no sooner than 8 ms across the longest cortico-cortical fibres. If two regions re-agree after a local perturbation faster than the synaptic route allows, the coupling has a non-synaptic part, and the timing measures the reach of the kernel tail of Section 4 directly.

The field’s strength sets the distance at which the test of the field’s causal cone in Section 6 can decide anything. Endogenous fields in cortex peak near 2.4 mV/mm in vivo (Fröhlich & McCormick, 2010), networks of neurons detect fields from 0.14 mV/mm (Francis et al., 2003), and the potential of a local cortical source decays as $r^{-2.1}$ across millimetres (Rebollo et al., 2021). With the peak placed at the nearest distance of that measurement, the field of a local source falls to the detection threshold within 3.8 mm, and misses it by 5 orders of magnitude at 15 cm and by 3.5 at 5 cm. The field’s cone is therefore tested at millimetre distances, where activity has been observed to synchronize across a cut through cortex by the field alone (Rebollo et al., 2021). At centimetre distances P predicts that re-agreement arrives through synaptic coupling, which is graded as well.

The timing can be measured with intracortical recording, with care. The stimulation artefact saturates clinical amplifiers during the first milliseconds, masking the earliest monosynaptic responses (Keller et al., 2014), and early responses at distant contacts carry a volume-conducted component that

scales with distance as the artefact does (Prime et al., 2020). Three measures separate active re-agreement from the passive field that carries the stimulus: template subtraction of the artefact (Trebaul et al., 2016); multi-unit activity or current-source density at the distant site, which the passive spread of the local field potential does not reach (Kajikawa & Schroeder, 2011); and a deadline set for each subject from the tractography length of the connecting fibres (Caminiti et al., 2013).

J Further related work

Biology and the medium. Seth (2026) argues that consciousness may depend on biological properties that computation does not capture. P locates the relevant property in how the substrate is organised, and admits non-biological hardware that has it (Section 5). Mortal computation, whose parameters are useful only on the hardware instance that learned them (Hinton, 2022; Ororbia & Friston, 2023), shares P’s refusal to separate the computation from the medium.

Analog and digital. On the mechanistic account of Piccinini (2008), a digital computer manipulates strings of digits, states its components distinguish reliably, and which class a system belongs to is fixed by its mechanism. P’s symbol, a content read through a noise margin, is that reliability stated as a property of the readout, with the margin measured in the parts’ own noise; P then asks what the reliability costs, namely graded dependence below the margin. Maley (2011) separates analog from continuous and digital from discrete representation, and P’s line follows the second pair, not the first: a spike time is a physical content whether or not it represents by analogy, and a continuous quantity latched through a margin is a symbol. Maudlin (1989) argues that a computation’s counterfactual structure can be carried by machinery that is idle on the actual run, which sits badly with consciousness

supervening on present physical activity. P states its counterfactual condition on present activity: graded dependence asks how the actual carriers respond to changes of the actual state, above their noise, and idle machinery responds to none.

Field theories. The electromagnetic field theory of McFadden (2020) identifies consciousness with the brain’s field, supported by evidence that endogenous fields entrain cortical activity (Fröhlich & McCormick, 2010). Miller et al. (2026) place cognition and consciousness in travelling waves that synaptic and ephaptic coupling sustain together, and take consciousness to arise when those waves bring cortex into a globally integrated state. Their analog computation is set apart from digital computation by running in parallel. P treats the field, and the waves it carries, as ways among several to realize graded coupling, and draws the line between analog and digital at the noise margin, which a parallel digital machine keeps.

K Formal results

Each formal result the paper states is a theorem of the Lean 4 development in the project repository, in its four `Phase10` modules. Table 1 maps each result to its declaration and to the section that argues from it. Every theorem rests only on Lean’s foundational axioms `propext`, `Classical.choice` and `Quot.sound`, which the build verifies for every declaration of the development. Small concrete models in `Examples/` meet the hypotheses of the bounds and show where they are attained.

Table 1: The paper’s formal results and the Lean declarations that carry them.

Result	Section	Declaration
A readout with a margin, under continuous micro dynamics, has no graded dependence on any region	3	<code>not_physicallyEnforced_of_margin</code>
The same for any content read from a finite word of thresholded continuous quantities	3	<code>not_physicallyEnforced_of_bits</code>
A thresholded continuous quantity has a margin off its threshold	3	<code>hasMargin_threshold</code>
A category read from a carrier depends gradedly on a region only where the carrier does	3	<code>GradedDependence.of_comp</code>
Every function of a bit word, after steps that keep the bits off their thresholds, depends gradedly on no region	3	<code>not_gradedDependence_iterate_of_bits</code>
Shells in series bound the effective conductance	A	<code>conductance_le_shell</code>
Crossing coupling at most $c(k+1)$ gives resistance at least $\log(n+1)/c$	A	<code>conductance_mul_log_le</code>
An averaging chain from a to b bounds the resistance by $2(\sum_k \rho_k)^2$	A	<code>one_le_conductance_of_chain</code>
A tail $K(r) \gtrsim r^{-(2+\sigma)}$ with $\sigma < 2$ bounds the conductance below at every distance	A	<code>le_conductance_of_powerLaw</code>
Adding coupling never raises the effective resistance	A	<code>conductance_mono</code>
A direct coupling caps the resistance at its reciprocal	A	<code>le_conductance_of_edge</code>

Result	Section	Declaration
A Lipschitz encoder carries mean-square phase agreement to content agreement	4	<code>integral_dist_sq_le</code>
A relative encoder reports one content at every site of a wave	A	<code>relContent_eq_of_wave</code>
A relative encoder's discrepancy is bounded by the departure from a wave	A	<code>dist_relContent_le</code>
Tracking a remote variable requires it to lie in the causal past	4	<code>mem_ball_of_reconstructs</code>
Contraction by κ reaches ε by step $\log(d_0/\varepsilon)/\log(1/\kappa)$	4	<code>disc_iterate_le_of_log_le</code>
A discrepancy shrinking by at most ρ per step is above ε before $\log(d_0/\varepsilon)/\log(1/\rho)$	4	<code>lt_disc_iterate_of_lt_log</code>
Graded dependence of a bit word requires some bit exactly on its threshold	5	<code>exists_threshold_of_gradedDependence</code>
A one-region change below $\delta/(LL_v)$ changes no bit	5	<code>bits_eq_of_lipschitz</code>
Under noise, a one-region change below r/L moves the probability of every event of the next content by at most the error rate	3	<code>law_le_of_marginRate</code>
The same for anything computed from noisy bits, with error rate the probability of landing within δ of a threshold	3	<code>bits_law_le_of_lipschitz</code>
The response of a content to one region grows with the scale	3	<code>response_mono</code>

Result	Section	Declaration
Below r/L the response is at most the error rate	3	<code>response_le_of_marginRate</code>
Graded dependence above the noise floor at every overlap holds exactly when every overlapping pair is one system	3	<code>enforcedAtNoiseFloor_iff</code>
A region whose content has a margin with error rate below the fluctuation is coupled to no region, and is one system only with itself	3	<code>not_coupledAtNoiseFloor_of_record, sameSystem_iff_eq_of_record</code>
For any set of such records, two regions are one system exactly when a chain of couplings avoiding every record joins them	3	<code>sameSystem_iff_of_records</code>
A content that depends on a region only through a record's symbol responds to it by at most the record's error rate, so write-back through a record couples nothing	3	<code>shiftResponse_le_of_factors, not_coupledAtNoiseFloor_of_factors</code>
A record that reads two graded regions joins neither, a write-back through it links neither, and a graded reader under linear dynamics with uniform noise joins two uncoupled regions	3	<code>tally_sameSystem_iff, writeBack_not_coupled, reader_sameSystem</code>
Asking graded dependence above the noise floor of every carrier implies asking it of some carriers of each region	3	<code>EnforcedAtNoiseFloor.enforcedByCarriers</code>

Result	Section	Declaration
A state at which some carriers of each region pass and not every carrier does	3	spare_ enforcedByCarriers, spare_not_ enforcedAtNoiseFloor
Regions joined by carriers at every overlap are one system	3	EnforcedByCarriers. regionSameSystem
Bits with error rate below the fluctuation: no carrier of any region reaches any other, and each region is a system of its own	3	not_ reachesAtNoiseFloor_of_ bits, regionSameSystem_ iff_eq_of_bits
With K independent carriers each passing within a window with probability at least p , two regions go unlinked in the window with probability at most $(1 - p)^K$	3	measure_not_ reachesWithin_le
Without independence, two regions go unlinked in the window with probability at most the variance of the number of passing carriers over the square of its mean	3	measure_not_ reachesWithin_le_ variance
Bits with error rate below the fluctuation couple no pair of regions, so each region is a system of its own	3	sameSystem_iff_eq_of_ bits
The same bits fail graded dependence above the noise floor at any overlap of two regions	3	not_ enforcedAtNoiseFloor_ of_bits
A smaller shift of a Gaussian moves no event more than a larger one	3	gaussian_shift_le

Result	Section	Declaration
Where the response grows with the change, graded dependence above the noise floor is decided by a change of one noise amplitude	3	<code>gradedAboveNoise_iff_of_mono</code>
Under linear dynamics with Gaussian noise, the response grows with the change, and graded dependence and coupling are decided at one noise amplitude	3	<code>gaussLin_coupledAtNoiseFloor_iff</code>
A shift of a density with a single peak moves no event more than a larger shift, in either direction	3	<code>lawRise_map_add_mono_of_unimodal</code>
Under linear dynamics with additive noise whose density has a single peak, graded dependence and coupling are decided at one noise amplitude	3	<code>noisyLin_coupledAtNoiseFloor_iff</code>
Laws with a monotone likelihood ratio, exponential families among them, cross once	C	<code>singleCrossing_of_mlr, singleCrossing_expFamily</code>
Where the next content's laws cross once along a parameter the region moves monotonically, graded dependence above the noise floor is decided by a change of one noise amplitude up and one down	C	<code>gradedAboveNoise_iff_of_crossing</code>